Math 2C03 - Assignment 4

Fall 2026

This is one generated instance of the assignment. Your Childsmath questions may have different numerical values or choice orders. Submit your answers in Childsmath.

Question 1

Let

$$L[y] = a_n(x)y^{(n)}(x) + a_{n-1}(x)y^{(n-1)}(x) + \cdots + a_1(x)y'(x) + a_0(x)y(x),$$

where $a_0(x), a_1(x), \ldots, a_n(x)$ are given functions. Consider the nth order linear differential equation

$$L[y] = (8 + 6x + 8x^2)e^{3x}. \qquad (*)$$

If it is known that

$$\begin{aligned} L[y_1(x)] &= (-4 - 4x^2)e^{3x} \quad \text{when } y_1(x) = (2 - 3x + 3x^2)e^{3x}, \\ L[y_2(x)] &= -3xe^{3x} \quad \text{when } y_2(x) = (-4 + 2x - 4x^2)e^{3x}, \end{aligned}$$

find a particular solution, $y_p(x)$, to $(*)$.

Question 2

Find the two values of m so that the function $y = x^m$ is a solution to the following differential equation.

$$x^2y'' + 2xy' - 72y = 0.$$

Separate your answers with a comma. If there is only one value of m then write it twice, e.g., $12, 12$.

Question 3

Consider the following statements.

(i) $\{2e^{-5x}, 5xe^{-5x}\}$ forms a fundamental set of solutions of $y'' + 10y' + 25y = 0$ on $(-\infty, \infty)$.

(ii) $\{x^{1/2}, x^{9/2}\}$ forms a fundamental set of solutions of $x^2y'' - 4xy' + \frac{9}{4}y = 0$ on $(0, \infty)$.

Determine which of the above statements are True (1) or False (2).

So, for example, if you think that the answers, in the above order, are True, False, then you would enter '1,2' into the answer box below (without the quotes).

Question 4

Solve the following initial value problem.

$$x^2y'' + 21xy' + 104y = 0, \qquad y(1) = 7, \quad y'(1) = 6.$$

Question 5

Consider the following ordinary differential equation:

$$y^{(5)}(x) + 2y^{(4)}(x) + 7y'''(x) = 2x^2 + 3xe^{-x}\cos(\sqrt{6}\,x).$$

If you were to use the Method of Undetermined Coefficients, which form should you use in order to be guaranteed to be able to find unique coefficients yielding a particular solution?

(A)
$$x^3(Ax^2 + Bx + C) + x(D + Ex)e^{-x}\cos(\sqrt{6}\,x) + x(Fx + G)e^{-x}\sin(\sqrt{6}\,x)$$

(B) $x^3(Ax^2 + Bx + C) + x(D + Ex)e^{-x}\cos(\sqrt{6}\,x)$

(C) $Ax^5 + Bx^2e^{-x}\cos(\sqrt{6}\,x) + Cx^2e^{-x}\sin(\sqrt{6}\,x)$

(D)
$$x^3(Ax^2 + Bx + C) + x(D + Ex)e^{-x}\cos(\sqrt{6}\,x) + x(Dx + E)e^{-x}\sin(\sqrt{6}\,x)$$

(E)
$$(Ax^2 + Bx + C) + (D + Ex)e^{-x}\cos(\sqrt{6}\,x) + (D + Ex)e^{-x}\sin(\sqrt{6}\,x)$$

(F) $Ax^2 + Bxe^{-x}\cos(\sqrt{6}\,x) + Bxe^{-x}\sin(\sqrt{6}\,x)$

(G) $(Ax^2 + Bx + C) + (D + Ex)e^{-x}\sin(\sqrt{6}\,x)$

(H) $x^3(Ax^2 + Bx + C) + x(D + Ex)e^{-x}\sin(\sqrt{6}\,x)$

Question 6

Solve the following initial value problem.

$$y'' - 6y' + 8y = x + e^{4x}, \qquad y(0) = 0, \quad y'(0) = 3.$$

Question 7

Solve the following initial value problem.

$$y^{(4)} - 6y''' + 5y'' = x,$$
$$y(0) = 0, \qquad y'(0) = 0, \qquad y''(0) = 0, \qquad y'''(0) = 0.$$

Question 8

For which of the cases below is the given two-parameter family of functions the general solution of the corresponding nonhomogeneous differential equation on the indicated interval?

(i) $y'' - 7y' + 12y = 3x^2 - 42x + 6$;

$$y = c_1 e^{8x} + c_2 e^{3x} + 3x^2, \qquad (-\infty, \infty).$$

(ii) $y'' - 12y' + 36y = 210 \sin x - 72 \cos x$;

$$y = c_1 e^{6x} + c_2 x e^{6x} + 6 \sin x, \qquad (-\infty, \infty).$$

(iii) $x^2 y'' - 16xy' + 72y = \frac{148}{x}$;

$$y = c_1 x^8 + c_2 x^8 + \frac{1}{x}, \qquad (0, \infty).$$

(A) all of them	(B) none of them
(C) (i) and (iii) only	(D) (iii) only
(E) (ii) only	(F) (ii) and (iii) only
(G) (i) and (ii) only	(H) (i) only

Question 9

Use the method of variation of parameters to find a particular solution to the following differential equation.

$$y'' - 6y' + 9y = \frac{e^{3x}}{25 + x^2}.$$

Question 10

Find a particular solution to the following differential equation using the method of variation of parameters.

$$x^2y'' - 11xy' + 35y = x^2 \ln x.$$